

**AMENDMENT NO. 3 NOVEMBER 2007
TO
IS 1554 (PART 1) : 1988 SPECIFICATION FOR
PVC INSULATED (HEAVY DUTY) ELECTRIC
CABLES**

**PART 1 FOR WORKING VOLTAGES UP TO AND
INCLUDING 1 100 V**

(Third Revision)

[Page 2, clause 6.1(b) (see also Amendment No. 1)] — Substitute
'Galvanized steel formed wire (strip), or' for 'Galvanized steel strip, or'.

[Page 2, clause 6.2, line 1 (see also Amendment No. 1)] — Substitute the
following for the existing matter:

'The galvanized round steel wires/formed steel wires (strips) used for armouring
shall conform to IS 3975 : 1999†'

(Page 2, footnote marked †) — Substitute the following for the existing
footnote:

†Low carbon galvanized steel wires, formed wires and tapes for armouring of cables —
Specification (third revision).'

(Page 6, clause 13.3) — Insert the following at the end:

"The tolerance on nominal dimensions shall be as per IS 3975 : 1999. However,
for formed steel wires compliance shall be ensured only for dimensions 'A' &
'C'."

(Page 6, clause 13.5.2) — Insert the following new clause after 13.5.2:

**'13.6 The round steel wires/formed steel wires taken from the cable shall meet
the following requirements:**

- a) The tensile strength of round steel wire/formed steel wire shall be not
less than 250 N/mm^2 and not more 580 N/mm^2 .
- b) The Elongation at break of round steel wire/formed steel wire shall be
not less than 6 percent.

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- c) Round steel wire shall meet the requirements of torsion test. The gauge length between Vices and the minimum number of turns without break shall be as per Table 6 of IS 3975 : 1999.
- d) The zinc coating shall not show any cracks and shall not flake off on rubbing by the bare finger when the formed steel wire is subjected to winding test.
- e) The uniformity of zinc coating of round steel wire/formed steel wire shall comply with the requirements of IS 3975 : 1999 subject to the following:
 - The minimum number of dips shall be reduced by one half-minute dip
 - In case of formed wires, dip test is applicable only for the face
- f) The mass of zinc coating of round steel wire shall be not less than 95 percent of the mass specified in Table 2 of IS 4826 : 1979.

The mass of zinc coating of formed steel wire shall be not less than 95 percent of the mass specified in IS 3975 : 1999.

- g) The resistivity of the round steel wire/formed steel wire shall meet the requirements of IS 3975 : 1999.

[Page 8, clause 15.1, informal table, Sl No. (b)] — Substitute the following for the existing matter:

b) Tests for round steel wire/formed steel wire (strip) armour:

1) Dimensions	13.3 of IS 1554 (Part 1)	Part 36 of IS 10810
2) Physical tests on round/formed wire:		
i) Tensile strength	13.6(a) of IS 1554 (Part 1)	Part 37 of IS 10810
ii) Elongation at break	13.6(b) of IS 1554 (Part 1)	Part 37 of IS 10810
iii) Torsion test for round wires	13.6(c) of IS 1554 (Part 1)	Part 38 of IS 10810
iv) Winding test for formed wires	13.6(d) of IS 1554 (Part 1)	Part 39 of IS 10810

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v) Uniformity of zinc coating	13.6(e) of IS 1554 (Part 1)	Part 40 of IS 10810
vi) Mass of zinc coating	13.6(f) of IS 1554 (Part 1)	Part 41 of IS 10810
vii) Resistivity	13.6(g) of IS 1554 (Part 1)	Part 42 of IS 10810

(Page 8, footnote marked ‡) — Substitute the following for the existing matter

‘‡Low carbon galvanized steel wires, formed wires and tapes for armouring of cables — Specification (*third revision*).’

(ET 09)

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